Pyramids, prisms, and complete geometry of Conway closures

Abstract

We construct the complete hyperbolic metric of an augmented link specified by a completed Conway expression with two completed closing children. A strict stationary point of its positive Newton source determines ideal quadrangular pyramids and triangular prisms. Stationarity matches their entire ordered quadrilateral faces. We prove convexity throughout the assembly and identify the final colored boundary with the literal projection cut of the original diagram. Explicit face maps give the complete edge returns, including the half-twist partners, and a common horocycle decoration proves cusp completeness. The volume is twice the sum of the cell volumes, hence 2π times the stationary source action. We also compute the primitive crossing-circle lattice and give a constructive choice of half twists for which the main strands form a knot. No assertion about signed filling asymptotics or a nearby deformation is needed.

1 The diagrams, the action, and the geometric theorem

A crossing circle surrounds two strands and bounds a disk punctured twice by them. The augmented diagram retains this circle and removes full twists from the two strands. A retained half twist changes the identification of the two sides of the disk. Throughout this paper the disk, the endpoint order and the embedding of the whole construction in the projection sphere are part of the data.

Label the four endpoints of a two-string box in cyclic order by $0, 1, 2, 3$, starting at the bottom left and moving to the bottom right. For a stack FG, with F below G, join F_2 to G_1 and F_3 to G_0. Its exposed tuple is (F_0, F_1, G_2, G_3). A clockwise quarter-turn moves the entire box and replaces its tuple by (F_1, F_2, F_3, F_0). Cylindrical closure joins the exposed endpoints 0 to 3 and 1 to 2. These are physical operations on the diagram, including all crossing disks.

Definition 1.1. Let h be the quarter-turn of the elementary crossing-circle tangle, with no retained twist. A completed expression is generated by

$$E ::= hh \mid \mathsf{T}(E) \mid E_1E_2 \mid hE \mid Eh.$$

A closed expression is $C(FG)$, where both F and G are completed. Every occurrence of h is a distinct crossing circle.

This is Definition 2.1 of [1]. The adjective completed specifies a rule for the expression, rather than a conclusion about a metric. A single bare leaf at a closing child is outside the rule. Write H for the number of leaves. Choose $\varepsilon_\ell \in \{-1, 0, 1\}$ at each leaf. The augmented link L_E^ε retains the indicated half twist and all its crossing circles; put $Y_E^\varepsilon = S^3 \setminus L_E^\varepsilon$. At this stage the main strands may have several components.

We recall the real action with its normalization. Put

$$J(t) = \int_0^t \log(2 \sin \pi u)\, \mathrm{d}u, \qquad 0 \le t \le 1, \qquad J(1-t) = -J(t).$$

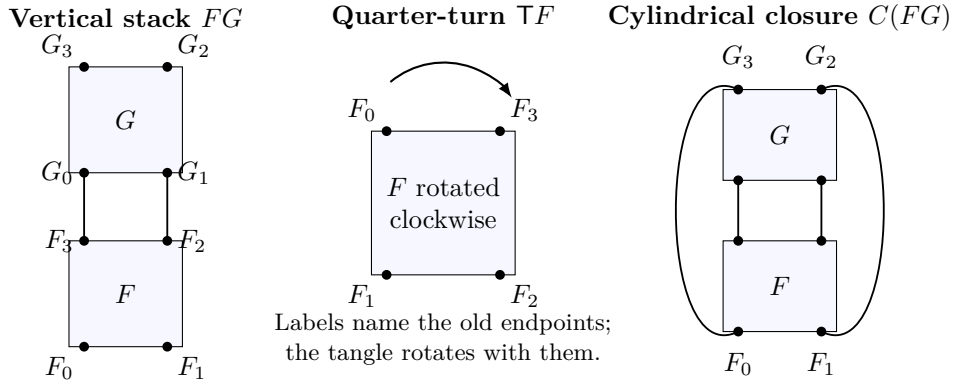


Figure 1: The ordered stack, physical clockwise turn and cylindrical closure. The label of an exposed endpoint is transported with its box. The two connections inside a stack and the two outside closing arcs are retained in the topological identification.

The leaf and product functions are

$$g(a) = -2J(a) - 2J(1/2 - a) = J(2a) - 4J(a), \tag{1.1}$$

$$m(x, y, z) = J(x + y) + J(z) - J(x) - J(y) - J(z - x) - J(z - y) - J(x + y - z). \tag{1.2}$$

A leaf has $0 < a < 1/2$. A strict product has

$$\max(x, y) < z < x + y < 1, \qquad x, y > 0. \tag{1.3}$$

It is useful to write

$$p = z - x, \quad q = z - y, \quad r = x + y - z, \quad w = 1 - x - y. \tag{1.4}$$

All four numbers are positive, and

$$p + q + 2r + w = 1, \quad x = q + r, \quad y = p + r, \quad 1 - z = r + w.$$

A physical turn replaces an output parameter b by $1 - b$. At cylindrical closure the two outputs are b and $1 - b$. The closing gate has been eliminated using that relation; it is not an additional volume term. The action $\mathcal{F}$ is the sum of g over leaves and m over product vertices on the resulting polytope. Write F_0 for its maximum.

The precise variational input is as follows. Proposition 5.3 of [1] proves that F_0 is attained at a unique strict point and that its Hessian in the free wire coordinates is negative definite. It also proves that the fixed-output profile $f_E(b)$ of a completed box is real analytic for $0 < b < 1$. The profile maximizes over its internal wires, so the envelope rule computes its derivative from its last cell. Lemma 6.1 of [1] gives, at the closed maximum,

$$\frac{1}{4} < a_\ell < \frac{\arctan 2}{\pi}, \qquad \frac{1}{2} < \cot(\pi a_\ell) < 1. \tag{1.5}$$

Only the first two conclusions are needed to construct the metric. The last will bound the crossing-circle shapes.

Theorem 1.2. *For every fixed closed expression in Definition 1.1 and every choice of retained half twists ε, the original exterior Y_E^ε has a complete finite-volume hyperbolic metric. The metric is assembled from two copies of a convex right-angled ideal polyhedron, and*

$$\mathrm{Vol}(Y_E^\varepsilon) = 2\pi F_0. \tag{1.6}$$

At the ℓth crossing circle the original meridian and disk longitude are a primitive ordered basis. With the half-twist convention of Section 7, the longitude-to-meridian ratio is

$$\tau_{\ell,\varepsilon} = \frac{2\mathrm{i}c_\ell}{1-\varepsilon_\ell \mathrm{i}c_\ell}, \qquad c_\ell = \cot(\pi a_\ell) \in (1/2, 1). \tag{1.7}$$

All main cusp ends are complete as well.

The conclusion identifies the complement of the specified diagram, not just a manifold with the same cell counts. It requires no hyperbolic polyhedron existence theorem or assertion about the complete character variety. The ideal-tetrahedron volume formula is the classical input recalled in Section 2. The Newton stationary-point theorem is the only imported construction-specific existence result.

Corollary 1.3. *Let $Z_{F,G,N}$ be the pure complete auxiliary-color source of [1, Section 2], with each leaf evaluated as h_N and no retained half twist. It is strictly positive for all sufficiently large N, and through all sufficiently large integer levels*

$$\lim_{N\to\infty} \frac{2\pi}{N} \log Z_{F,G,N} = \mathrm{Vol}(Y_E^0) = \mathrm{Vol}(Y_E^\varepsilon).$$

Proof. Corollary 5.4 of [1] gives the limit F_0 before multiplying by 2π; Theorem 1.2 identifies that number with the metric volume. The scalar here is the pure positive source, not a signed filled-knot invariant. □

The decomposition of fully augmented links into right-angled polyhedra and the effect of half twists are established parts of augmented-link geometry; see [3]. The relation between a concave action and hyperbolic volume belongs to the variational setting discussed by Rivin [4]. The task here is to identify the particular stationary pyramids and prisms with the original completed diagram, with all orders and peripheral markings retained. We give that construction directly.

2 The two hyperbolic cells

We work in upper half-space, with ideal boundary $\mathbb{C} \cup \{\infty\}$. A geodesic plane is vertical over a line or is a hemisphere orthogonal to the boundary plane. A horosphere centered at ∞ is horizontal; its induced metric is Euclidean up to scale. An ideal tetrahedron with dihedral angles $\pi a, \pi b, \pi c$ at an ideal vertex, where $a, b, c > 0$ and $a + b + c = 1$, has volume

$$\begin{aligned} V(a,b,c) &= \Lambda(\pi a) + \Lambda(\pi b) + \Lambda(\pi c) = -\pi[J(a) + J(b) + J(c)], \\ \Lambda(t) &= -\int_0^t \log(2\sin u)\,\mathrm{d}u. \end{aligned} \tag{2.1}$$

This is the ideal-tetrahedron formula of [2]. Its normalization, including its vanishing at a flat tetrahedron, fixes the absolute volume used below.

2.1 The elementary pyramid

For a leaf parameter a, put $E = e^{2\pi \mathrm{i} a}$ and take

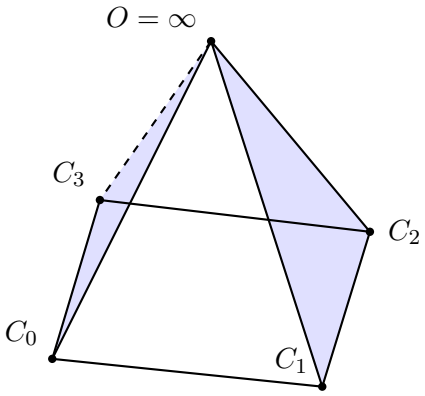
$$O = \infty, \qquad (C_0, C_1, C_2, C_3) = (1, E, -1, -E). \tag{2.2}$$

Their convex hull is an ideal quadrangular pyramid. The four finite vertices form a rectangle, so the four edges incident to O have dihedral angle $\pi/2$. Its base is an ordered quadrilateral

gluing face. Call its edges $01, 23$ free and $12, 30$ cap edges. The base angles are πa at the free edges and $\pi(1/2 - a)$ at the cap edges. These follow by intersecting the vertical side planes with the unit hemisphere, or by reading the half-arcs of the unit circle.

Color (O, C_1, C_2) and (O, C_3, C_0) shaded. The other two side triangles are white. Shaded faces will record the sides of a half crossing disk; white faces will record regions of the projection sphere. A base diagonal gives two positive tetrahedra with angle triple $(a, 1/2 - a, 1/2)$. Consequently

$$\text{Vol(pyramid)} = 2V(a, 1/2 - a, 1/2) = \pi g(a). \tag{2.3}$$



quadrilateral gluing face

Ideal hyperbolic pyramid.

All five vertices are ideal. The shaded triangular faces are (O, C_1, C_2) and (O, C_3, C_0).

In the boundary plane: $C_0 = 1$, $C_1 = e^{2\pi i a}$, $C_2 = -1$, $C_3 = -e^{2\pi i a}$.

Figure 2: The ideal hyperbolic pyramid of one leaf. The two shaded triangles retain their labels throughout the construction. The ordered quadrilateral base is auxiliary cut data, rather than the four-punctured boundary of the physical tangle box.

2.2 The multiplication prism

Given a strict product, use (1.4) and put

$$v_0 = 1, \quad v_1 = e^{2\pi i p}, \quad v_2 = e^{2\pi i (p+r)}, \quad v_3 = e^{2\pi i (p+r+q)}. \tag{2.4}$$

The successive half-arcs are $p, r, q, r + w$. Let P be the intersection of $v_0 v_1$ and $v_2 v_3$. The triangle (P, v_1, v_2) has angles $\pi w, \pi y, \pi x$ at P, v_1, v_2, respectively. The sine rule gives

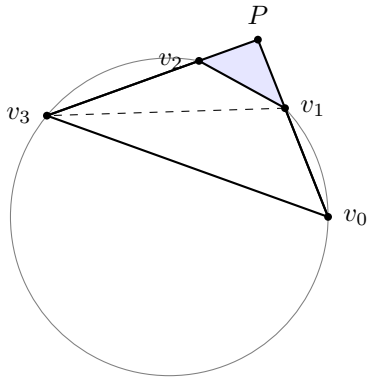
$$\begin{aligned} P &= v_1 + \frac{\sin \pi x \sin \pi r}{\sin \pi w \sin \pi p}(v_1 - v_0) \\ &= v_2 + \frac{\sin \pi y \sin \pi r}{\sin \pi w \sin \pi q}(v_2 - v_3). \end{aligned} \tag{2.5}$$

Both coefficients are positive. Thus P lies beyond v_1 from v_0 and beyond v_2 from v_3, and $|P| > 1$.

Start with the quadrangular pyramid $(\infty, v_0, v_1, v_2, v_3)$ and attach (∞, P, v_1, v_2) on the other side of their common triangle. The faces over $v_0 v_1$ and $P v_1$ are coplanar, as are those over $v_3 v_2$ and $P v_2$. After merging them the boundary consists of two white triangles (P, v_1, v_2) and (∞, v_0, v_3) and three ordered quadrilaterals

$$F_x = (\infty, v_0, v_1, P), \quad F_y = (\infty, v_3, v_2, P), \quad F_z = (v_0, v_1, v_2, v_3). \tag{2.6}$$

The vertical supporting planes lie over $v_0 v_1$, $v_2 v_3$ and $v_3 v_0$. The other two planes are the hemispheres through (v_0, v_1, v_2, v_3) and (P, v_1, v_2). Formula (2.5) puts P outside the first



Boundary-plane construction. $P = (v_0v_1) \cap (v_2v_3)$ lies outside the unit circle. The sixth ideal vertex is ∞. The three quadrilateral gluing faces have ordered vertices (∞, v_0, v_1, P), (∞, v_3, v_2, P) and (v_0, v_1, v_2, v_3).

$p = .12, \quad r = .10, \quad q = .17, \quad w = .51$

Figure 3: The boundary-plane support data for a multiplication prism. The point P is the exact intersection of the two extended chords; the sixth ideal vertex is ∞. The example illustrates one strict tuple. Formula (2.5) proves the construction throughout (1.3).

boundary circle and v_0, v_3 outside the second. Their inward half-spaces therefore contain precisely the attached pieces. This proves that their union is a convex ideal triangular prism.

The full dihedral-angle table is

Edges	Angle divided by π
$Pv_1,\ \infty v_0$	x
$Pv_2,\ \infty v_3$	y
$v_1v_2,\ v_0v_3$	$1-z$
$P\infty$	w
v_0v_1	p
v_2v_3	q

The first two rows are the angles of (P, v_1, v_2) and their opposite edges; the last two are half-arcs of the unit circle. At each trivalent ideal vertex the interior dihedral angles sum to π; for instance the three at P sum to $\pi(x + y + w) = \pi$.

Triangulate the quadrangular pyramid across v_0v_2. Its two tetrahedra have triples $(p, r, 1-p-r)$ and $(p+r, q, r+w)$. The complementary terms with arguments $p+r$ and $1-p-r$ cancel in (2.1). The remaining four Lobachevsky terms also equal the sum for the two positive triples $(p, q, 2r+w)$ and $(p+q, r, r+w)$. The attached tetrahedron has triple (y, x, w). Hence

$$\begin{aligned} \mathrm{Vol}(\text{prism}) &= V(y, x, w) + V(p, q, 2r+w) + V(p+q, r, r+w) \\ &= \pi m(x, y, z). \end{aligned} \tag{2.7}$$

All these triples are strictly positive and sum to one. The equality is an ordinary sum of metric volumes.

3 Ordered face matching and convex assembly

3.1 The whole quadrilateral, including its shear

For cyclically ordered points on a circle or line define

$$\varkappa(a,b,c,d) = \frac{(a-b)(c-d)}{(a-d)(b-c)} > 0, \tag{3.1}$$

with limits at infinity. Sending three ordered vertices to $(\infty, 0, 1)$ shows that $\varkappa$ determines the ordered ideal quadrilateral up to isometry. It fixes its diagonal shear as well as its labels. A one-step cyclic shift, or reversal fixing the first vertex, inverts $\varkappa$. Full reversal $(a,b,c,d) \mapsto (d,c,b,a)$ preserves it.

The prism coordinates give

$$\varkappa(F_x) = \frac{\sin \pi x \sin \pi r}{\sin \pi w \sin \pi p} = e^{-m_x}, \qquad \varkappa(F_y) = \frac{\sin \pi y \sin \pi r}{\sin \pi w \sin \pi q} = e^{-m_y}, \tag{3.2}$$

$$\varkappa(F_z) = \frac{\sin \pi p \sin \pi q}{\sin \pi z \sin \pi r} = e^{-m_z}. \tag{3.3}$$

Differentiating (1.1) gives the leaf formula $\varkappa(C_0, C_1, C_2, C_3) = \tan^2(\pi a) = e^{-g'(a)}$. For a product FG the actual attachment maps are

	C_0	C_1	C_2	C_3
F	v_0	v_1	P	∞
G	∞	P	v_2	v_3.

(3.4)

The first is a cyclic shift of F_x, and the second is reversal fixing the first vertex of F_y. They invert the input value of $\varkappa$. If f_F is the stationary output profile of the lower child, its wire equation is $f_F'(x) + m_x = 0$; the upper equation is the same with G, y. Thus the output and input values of $\varkappa$ multiply to one, exactly the condition for the assignments (3.4) to extend to whole-face isometries. The new profile has derivative m_z. This inductively proves the output formula $\varkappa = e^{-f_E'}$. A turn cyclically relabels the face and replaces b by $1 - b$, which has the same reciprocal effect.

3.2 Why successive attachments stay convex

Local angle equations do not by themselves exclude overlap of distant pieces. The following elementary argument controls the whole union.

Lemma 3.1. *Place two finite ideal convex polyhedra on opposite sides of the plane of an isometric maximal face. If every finite boundary seam of their union has total interior angle at most π, the union is convex. If the common face is a quadrilateral with strictly nonflat boundary edges in each piece, their ideal closures have no intersection other than the four recorded face vertices.*

Proof. The finite part of the union is closed in hyperbolic space and locally convex. At a seam edge this is exactly the angle condition; away from a seam it follows from convexity of the pieces. Any two finite points can be joined in the union by a finite-length path. A minimizing sequence with bounded length stays in a compact ambient hyperbolic ball. Constant-speed parametrization and a uniformly convergent subsequence give a shortest path in the union. Local convexity makes it an ambient geodesic near every point. It is therefore the unique ambient segment, proving convexity.

The opposite closed half-spaces meet only in their supporting plane. Their ideal closures can meet only on its boundary circle. The intersection of either polyhedron with that plane

is exactly its maximal face. An additional ideal vertex in the plane would enlarge the face across a boundary edge and make that edge flat, contrary to the strict hypothesis. In the Klein model the closure of an ideal quadrilateral meets the sphere at infinity at precisely its four vertices. This excludes every unrecorded ideal intersection. □

An elementary output has free angle πa and cap angle $\pi(1/2-a)$. Inductively, a completed output b has free angle πb and cap angle $\pi(1-b)$. Attach the lower child first. At a cap seam a leaf gives total angle $\pi/2$, while a completed child gives π and joins two white faces. The edge shared by the two prism inputs now has angle $\pi(w+x)=\pi(1-y)<\pi$. Thus the other input remains a maximal supported quadrilateral face. Lemma 3.1 applies again when the upper child is attached. We have not assumed nonintersection between two independently placed children.

The final common input edge has angle $\pi(w+x+y)=\pi$. The free output seams have angles $\pi(p+x)=\pi z$ and $\pi(q+y)=\pi z$; the cap output edges have angle $\pi(1-z)$. Every remaining edge is an original shaded-triangle edge and is right angled. Flat seams join white faces. Strictness of the output parameter keeps its face maximal and supported, so the induction continues.

There is a separate combinatorial check at ideal vertices. At every open stage the four exposed corners are four unpaired original terminal points. Product consumes the disjoint pairs F_2/G_1 and F_3/G_0 and never exposes them again; a turn merely permutes the unpaired points. Later identifications cannot extend a consumed pair into a longer chain. Each final main vertex is exactly a pair of original terminals, and a leaf's four corners stay distinct. Its crossing-circle apex is not among those terminals. Together with Lemma 3.1, this shows that every leaf retains two distinct shaded triangles sharing only their original apex.

3.3 Closing the two completed children

At the root the outputs are $b, 1-b$. The free root equation is $f'_F(b)=f'_G(1-b)$, so their ordered quadrilaterals have the same $\varkappa$. The closing map is

$$C_j^F \longmapsto C_{3-j}^G, \qquad j \pmod 4. \tag{3.5}$$

Full reversal preserves $\varkappa$ and simultaneously realizes the two stack connections and the two outside closing connections. Each pair of free angles sums to π, as does each pair of cap angles. The last gluing thus joins white faces across flat seams. The result is a convex right-angled ideal polyhedron P_E, with

$$\mathrm{Vol}(P_E)=\pi\sum_{\ell} g(a_\ell)+\pi\sum_{\text{products}} m(x,y,z)=\pi F_0. \tag{3.6}$$

There are $2H$ shaded triangles, $3H$ ideal vertices and $6H$ edges; Euler's formula gives $H+2$ white faces. These counts are consequences of the assembly. The next section identifies its entire labeled boundary.

4 Recovering the original cut balls

Place the pure main strands in the projection sphere, retaining the crossing disks in their disjoint elementary boxes. Cut the link exterior along that sphere. Each auxiliary circle becomes a semicircle in each projection ball, boundary-parallel along its original half crossing disk. Drilling a boundary-parallel semicircle introduces one handle; cutting along its half disk opens precisely that handle. The half disks are disjoint, so these cancellations can be performed successively in the original boxes. Each result is a ball with ideal vertices. Main

strand segments lie on its boundary; their removal creates ideal corners and no interior handles.

At a leaf the cut produces exactly (O, C_1, C_2) and (O, C_3, C_0) as its shaded triangles. The other faces are regions of the original projection sphere. We determine their full walks as follows. At O keep the cyclic order (C_0, C_1, C_2, C_3). At an outer corner each shaded triangle contributes the sector ordered by its next and previous neighbors. The identity strip joining two terminals concatenates their two sectors in planar order. A boundary walk entering a vertex along an oriented edge (u, v) exits along the neighbor preceding u in the cyclic order at v. Thus the shaded triangles and terminal pairings determine every remaining white walk.

The maps (3.4) identify F_2 with G_1 at P, and F_3 with G_0 at ∞. Their exposed order is $(F_0, F_1, G_2, G_3) = (v_0, v_1, v_2, v_3)$. The prism's two white triangles insert exactly the intervening white sectors. At a completed cap seam those sectors merge across a flat white edge; at a leaf cap seam their boundary remains an original shaded edge. Turns transport the cyclic order, and (3.5) realizes the original closing pairs. Hence the ribbon orders of the final nonflat graph are the orders of the literal projection cut. All faces, including every white walk, have the same labels and cyclic orders.

This gives a label-preserving cellular homeomorphism of each cut-ball boundary with ∂P_E. A sphere homeomorphism extends over a ball, for example radially after choosing ball coordinates. Gluing the two extended balls along their prescribed white and shaded faces reverses exactly the original cuts. The quotient is Y_E^0, with degree one and the original disk curves retained.

An ordered quadrilateral during assembly is auxiliary cut data. It is not the physical four-punctured boundary of the tangle box. We first used its modulus to assemble cells, and then used the closed ribbon identification to recover the physical complement. No boundary group representation is inferred from quadrilateral moduli alone.

For a retained half twist, cut the twice-punctured disk and rotate one side through π. This interchanges both punctures and the upper and lower half disks. The two balls and white faces are unchanged. Superscripts distinguishing the projection copies give the new shaded maps

$$\begin{array}{l|lll} S_0^+ \longrightarrow S_1^- & O & C_1 & C_2 \\ & O & C_3 & C_0 \\ \hline S_1^+ \longrightarrow S_0^- & O & C_3 & C_0 \\ & O & C_1 & C_2. \end{array} \tag{4.1}$$

The other maps are their inverses. Both signs give this unmarked pairing; the original meridian paths differ, as computed in Section 7. The same labeled balls therefore give the original Y_E^ε.

5 Entire face maps and edge returns

The construction matches complete faces isometrically. We now give the complex-coordinate dictionary for their identifications. It records the shear and checks the entire edge return, including translation. Matrices are projective, and composition acts from right to left.

5.1 The leaf and its adjacent white faces

Normalize the ordered base of a leaf to $(0, \infty, 1, R)$. At completeness,

$$E = e^{2\pi i a}, \qquad Z = \frac{1+E}{2}, \qquad R = \frac{(1+E)^2}{4E} = \cos^2(\pi a). \tag{5.1}$$

The apices in the two charts are Z and $Z/E = \overline{Z}$. Their white or completion maps are

$$W_0 = \begin{pmatrix} 1 & 0 \\ 0 & E \end{pmatrix}, \qquad W_1 = \begin{pmatrix} 1 & E-1 \\ 0 & E \end{pmatrix},$$
$$W_2 = \begin{pmatrix} E-R & R(1-E) \\ E-1 & 1-ER \end{pmatrix}, \quad W_3 = \begin{pmatrix} R & 0 \\ 1-E & ER \end{pmatrix}. \tag{5.2}$$

The elementary pyramid has actual white faces only at W_0, W_2. The expressions W_1, W_3 describe the maps supplied by the adjacent prism at its cap edges. This is proved below before using them in a group word.

The pure within-copy shaded map and the odd cross-copy maps are

$$S = \begin{pmatrix} 0 & -ER \\ 1 & -1-E \end{pmatrix}, \quad A = \begin{pmatrix} R & -R \\ 1 & ER-1-E \end{pmatrix}, \quad B = \begin{pmatrix} R/E-1-1/E & R \\ -1 & R \end{pmatrix}. \tag{5.3}$$

Their vertex assignments are

$$S : (Z, \infty, 1) \mapsto (Z, 0, R), \quad A : (Z, \infty, 1) \mapsto (Z/E, R, 0), \quad B : (Z, R, 0) \mapsto (Z/E, \infty, 1).$$

Three ordered vertices determine the maps; substitution of $Z^2 - (1+E)Z + ER = 0$ verifies their apex values. At the complete parameters all these matrices are invertible.

5.2 Compatibility across an entire prism

Use independent phase variables α, β, γ for $e^{2\pi i p}, e^{2\pi i r}, e^{2\pi i q}$. The world coordinates are

$$v_0 = 1, \quad v_1 = \alpha, \quad v_2 = \alpha\beta, \quad v_3 = \alpha\beta\gamma,$$
$$P = 1 + \alpha - \frac{\alpha[\alpha\beta(1+\gamma) - 1 - \alpha]}{\alpha^2\beta^2\gamma - \alpha}. \tag{5.4}$$

Indeed the chord lines have equations $P + \alpha\overline{P} = 1 + \alpha$ and $P + \alpha^2\beta^2\gamma\overline{P} = \alpha\beta(1+\gamma)$, which give (5.4). For an ordered quadrilateral (a, b, c, d) use

$$M_{a,b,c}(z) = \frac{(z-a)(c-b)}{(z-b)(c-a)}, \qquad R = M_{a,b,c}(d).$$

Its world white map is $\overline{M}^{-1} W_j(E, R) M$. Here the bar reciprocates α, β, γ; on the complete unit-circle chamber it is complex conjugation. For the lower, upper and output quadrilaterals in (3.4) and (2.6), their wire phases are $\beta\gamma, \alpha\beta, \alpha\beta\gamma$. In these common world coordinates the entire maps obey

$$W_{F,1} = W_{G,1} = W_{o,1} = W_{\text{top}}, \qquad W_{F,3} = W_{G,3} = W_{o,3} = W_{\text{bottom}},$$
$$W_{F,2} = W_{G,0}, \quad W_{o,0} = W_{F,0}, \qquad W_{o,2} = W_{G,2}. \tag{5.5}$$

The subscript o means output. The top and bottom maps take (P, v_1, v_2) and (∞, v_0, v_3) to their reflected triples. To verify (5.5), insert (5.4) into M and (5.2) and cancel the projective scalar. Each side takes the same three distinct indicated vertices to their reflected partners. This proves equality of whole maps, not just of their multipliers. The formulas also give nine rational identities in the three independent variables: three top, three bottom and the three remaining equalities.

Turns and closure have explicit compatibility as well. Put $T_Q(z) = (R-1)/(z-1)$ and $C_Q(z) = (z-R)/(z-1)$. Direct multiplication gives, with indices modulo four,

$$T_Q W_{j+1}(E, R) = W_j(E^{-1}, 1-R) T_Q,$$
$$C_Q W_j(E^{-1}, R) = W_{2-j}(E, R) C_Q. \tag{5.6}$$

Induction proves that all triangular pieces of a final white face have the same entire map. Every elementary completion expression W_1, W_3 is supplied by its parent prism. At the completed/completed root the closing seams merge white faces. Thus each W_j used at a leaf is an actual white-face map of the closed cut. Matching $\nsim$ alone would not establish this.

5.3 The full edge words

Write $\overline{S} = S(E^{-1}, R)$. The pure identities are

$$W_0 S = \overline{S} W_0, \qquad W_2 S = \overline{S} W_2, \qquad W_3 S = \overline{S} W_1. \tag{5.7}$$

For an odd leaf they are

$$W_0^{-1} B W_2^{-1} A = I, \quad W_2^{-1} B W_0^{-1} A = I, \quad W_1^{-1} B W_3^{-1} A = I. \tag{5.8}$$

Direct matrix multiplication proves the projective equalities: the first two odd products are $-R/E$ times I, and the last is $(R-1)/E$ times I. The reverse cycles are inverses. Every final edge is an original shaded-triangle edge, so this list covers all edge orbits. The incidence proof of Section 3 gives four distinct occurrences in each orbit. Each has angle $\pi/2$, and its full four-step return is the identity. The metric is smooth at every finite edge, with no axial translation hidden by the angle sum.

This is also a group check on the literal manifold. The dual two-skeleton of the original cut balls has a generator for each paired face and a relation for each edge cycle. Equations (5.7)–(5.8) are precisely those relations, with white maps already identified by (5.5). Thus the construction respects the original group presentation, rather than a presentation inferred from unrelated shape equations.

6 Cusp completeness and the absolute volume

Take two oppositely oriented copies of P_E. Pair their white faces by the projection labels. For a pure leaf, pair its shaded triangles within each copy, fixing O and sending $C_1 \mapsto C_0$, $C_2 \mapsto C_3$. For an odd leaf use (4.1). Section 4 identifies the quotient with Y_E^{ε}, and Section 5 proves smoothness away from the ideal vertices.

At each ideal vertex a horosphere cuts out a Euclidean rectangle. Its two opposite shaded sides have equal length. A unique choice of horosphere makes both lengths one. Make this choice independently at the finitely many ideal vertices. Each shaded ideal triangle then has three horocyclic corner segments of length one. An isometry of ideal triangles carrying corresponding vertices to corresponding vertices preserves precisely this decoration, including every permutation in (4.1).

The white maps preserve the decoration as well: they are the label-preserving maps between the two identical copies of the polyhedron, and the shaded-side normalization agrees at each matched vertex. Thus each cusp link is assembled from Euclidean rectangles without dilation. The original topological identification makes it a compact torus. It is a smooth Euclidean torus, since the edge returns have already excluded cone singularities.

A common shrinking factor makes all chosen horospheres disjoint and is preserved by all pairings. Beyond each link lies its complete horoball end. Their complement is compact, so the whole hyperbolic metric is complete. Equivalently the universal developing map is a complete local isometry to hyperbolic space and hence covers it. Every cell and cusp tail has finite volume. All attachments were unions with disjoint interiors, and the original cut has degree one. Therefore

$$\operatorname{Vol}(Y_E^{\varepsilon}) = 2 \operatorname{Vol}(P_E) = 2\pi F_0.$$

This proves the metric and volume assertions of Theorem 1.2. The equality is absolute: it follows by partitioning the actual metric into the displayed cells and using (2.1). No equality of volume derivatives or choice of an additive regulator constant is involved.

7 The original crossing-circle curves

The crossing-circle torus has two rectangles, one from each cut ball. The disk longitude follows the twice-punctured crossing disk. The meridian goes once around the auxiliary circle. Both are marked before the cuts. We trace their words and check primitiveness against the full Euclidean cusp area.

In the normalized chart (5.1) these words are

$$\mu_0 = S^{-1}, \qquad \lambda = W_2^{-1} W_0. \tag{7.1}$$

For λ, leave one disk side through white face 0 and return through face 2 with its inverse. For μ_0, cross the shaded identification once in the reverse direction. The two half-twist collars instead give

$$\mu_- = A^{-1} W_0, \qquad \mu_+ = W_0^{-1} B. \tag{7.2}$$

The first path passes through white face 0 and returns through A^{-1}; the second passes through B and returns through W_0^{-1}. These are the two half-disk rotations in the original disk frame. We label the first negative and the second positive. Equivalently the retained crossing R^{-1} uses the first collar and R uses the second. They give the same unmarked triangular pairing but differ by a full longitude in their meridian paths. This fixes the sign convention in (1.7) by peripheral paths.

Return to (2.2), where the auxiliary apex is ∞. The normalization is $M(z) = k(z-1)/(z-E)$ with $k = (1+E)/2$, and $M(\infty) = Z$. Conjugating the pure words by M gives

$$\mu_0 = E - 1, \qquad \lambda = -2(E+1), \qquad \mu_- = -2, \qquad \mu_+ = 2E \tag{7.3}$$

as translations in the original coordinate. For a cross-copy word the target normalization is the reflected one; composing the two maps in (7.2) returns to the original chart before this conjugation. The matrices (5.2) and (5.3) give (7.3) by direct multiplication. Consequently

$$\mu_\varepsilon = \mu_0 - \varepsilon\lambda/2 \tag{7.4}$$

as Euclidean vectors on the respective cusp developments. This compares the markings of different half-twist partners; it does not make half a longitude an integral curve on the pure torus.

The adjacent sides of the original rectangle have lengths $|E-1| = 2\sin\pi a$ and $|E+1| = 2\cos\pi a$. The two rectangles of the cusp have total area $8\sin\pi a\cos\pi a = 4\sin 2\pi a$. On the other hand

$$|\mathrm{Im}(\overline{\mu_0}\lambda)| = 4\sin 2\pi a. \tag{7.5}$$

The two traced closed curves span a sublattice of the actual cusp lattice; equality of areas makes its index one. Substitution of either odd meridian leaves the area unchanged, proving the primitive marking for those collars as well.

Finally $(E+1)/(E-1) = -\mathrm{i}\cot\pi a$. Equations (7.3)–(7.4) give (1.7), with $\mathrm{Im}\,\tau_\varepsilon = 2c_\ell/(1+\varepsilon^2 c_\ell^2) > 0$. The bound on c_ℓ is the imported leaf window (1.5). This completes Theorem 1.2. All main cusps were already included in the horocycle construction; no additional eigenvalue or deformation hypothesis is required.

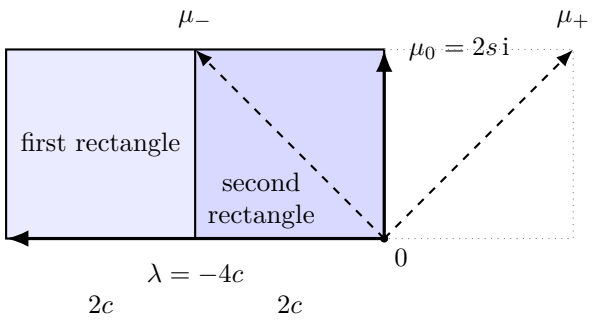


Figure 4: The crossing-circle cusp in the original leaf coordinate, after rotation by $e^{-\pi \mathrm{i} a}$. Put $s = \sin \pi a$ and $c = \cos \pi a$. The pure meridian is $2s\,\mathrm{i}$ and the disk longitude is $-4c$. Two rectangles of width $2c$ and height $2s$ give one fundamental parallelogram. The dashed vectors are $\mu_- = \mu_0 + \lambda/2$ and $\mu_+ = \mu_0 - \lambda/2$, each for its own half-twist partner.

8 Choosing a knot among the main components

The construction kept every main component. A finite endpoint argument produces a one-component choice throughout the stated grammar.

Proposition 8.1. *One can choose pure or odd leaves in every closed expression of Definition 1.1 so that its main strands form one component. Either sign may be used at an odd leaf. Subsequent full twists preserve that component count.*

Proof. Write the three pairings of four cyclically ordered endpoints as

$$I = (0,3)(1,2), \qquad U = (0,1)(2,3), \qquad X = (0,2)(1,3).$$

A pure leaf has pairing U, an odd leaf has pairing X, and turn exchanges I, U and fixes X. Tracing the two stack connections gives

lower/upper	I	U	X
I	I	U	X
U	U	U and one internal circle	U
X	X	U	I

Only $U \cdot U$ creates an internal component. The base hh has loop-free possibilities $\{I, U\}$. Put $A = \{I, U\}$, $B = \{I, X\}$, $C = \{U, X\}$, $D = \{I, U, X\}$. All products, discarding $U \cdot U$, give

	A	B	C
A	A	D	C
B	D	B	D
C	C	D	A

Each result contains at least two choices, also if an input already contains all three. Attaching a leaf is the row or column C. A turn permutes the possibilities. Inductively every completed box, and the root stack, has at least two loop-free choices. At least one is U or X, whose cylindrical closure is connected; the closure of I has two components. Backtracking the choices specifies the required leaves. Both signs have pairing X, and full twists have even endpoint permutation. □

The last assertion concerns connectivity. Theorem 1.2 proves hyperbolicity of the augmented exteriors, retaining the crossing circles. Hyperbolicity after filling those circles requires separate deformation and slope arguments.

9 Dependencies and finite algebra checks

The variational existence, uniqueness and leaf interval are the numbered inputs from [1] listed in Section 1. The cell and complement construction, entire face maps, cusp decoration and primitive marking are proved here. The ideal-tetrahedron volume formula (2.1) is classical background. The references on augmented links and variational geometry provide context and do not replace the ordered construction. No quantum asymptotic enters Theorem 1.2. Corollary 1.3 additionally uses the positive source limit of [1, Corollary 5.4].

The deterministic program `check.py` checks rational-function identities: nine prism maps, turn and closure compatibility, pure and odd edge returns, and the complete translations (7.3). It enumerates the endpoint pairings independently of the tables in Section 8. Its negative controls alter a vertex order and an odd meridian. The output records source hashes. These finite checks support the displayed algebra; they do not prove convexity, identify a manifold, establish completeness or supply the stationary-point theorem.

References

[1] *Newton coefficients and positive sources for Conway tangles*, companion paper, Definition 2.1, Proposition 5.3, Corollary 5.4 and Lemma 6.1. Companion PDF; source: `papers/newton-tangles/main.tex`.

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